

Confidently Wrong: Measuring and Mitigating Calibration Risks in LLMs for African Languages

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Abstract

Large language models (LLMs) are increasingly used in African languages, yet little is known about whether their confidence remains reliable. This creates a safety risk: models may become less accurate while remaining highly confident, increasing the likelihood of trusted misinformation. We evaluate four open-weight LLMs (**Qwen3-4B-Instruct**, **Gemma-4-12B**, **Aya-Expansive-8B**, and **AfroLlama-V1**) on African-language truthfulness and knowledge benchmarks (**Uhura-TruthfulQA** and **AfriMMLU**), measuring calibration using log-probability-based confidence estimates. We find that models become increasingly confident in incorrect answers as they move from English to African languages, creating a systematic overconfidence problem and a marked increase in expected calibration error. Among three mitigation strategies, temperature scaling proves most effective, reducing calibration error to near-English levels without retraining. Our findings suggest that African-language deployment risks stem not only from lower accuracy but also from models remaining confident when wrong, and that simple post-hoc recalibration offers a practical deployment-time safety intervention.

1. Introduction

LLMs are entering high-stakes, information-seeking use in African languages such as health, legal, educational, and civic information, often where users cannot easily verify an answer. The safety failure we study is a mismatch between confidence and correctness. When models remain highly confident despite being wrong, they can mislead users into treating incorrect information as reliable, which is particularly dangerous in high-stakes settings such as health, education, and civic information. A calibrated model, whose confidence tracks its accuracy, can instead *abstain*, declining or deferring when uncertain (Tomani et al., 2024). If calibration holds in English but breaks in African languages, the populations least served by current models are most exposed to confident misinformation. We investigate whether models become increasingly overconfident in African languages, quantify the extent of this misalignment between confidence and correctness, and evaluate practical strategies for mitigating it.

Our main contributions are:

1. An open-weight, reproducible calibration audit of four LLMs on African-language truthfulness and knowledge benchmarks, using answer-text log-probabilities to obtain robust confidence estimates for instruction-tuned models.
2. Evidence that calibration degrades substantially from English to African languages, revealing systematic overconfidence and a growing mismatch between confidence and correctness across models and tasks.
3. An evaluation of practical mitigation strategies, including temperature scaling, abstention, and few-shot prompting, together with a deployment framework showing that lightweight post-hoc recalibration is an effective safety intervention for African-language LLMs.

2. Related Work

Neural networks are often overconfident, and temperature scaling is a standard single-parameter post-hoc calibration method (Guo et al., 2017); for free-form generation, sampling-based methods such as semantic entropy estimate uncertainty without labelled data (Kuhn et al., 2023). Tomani et al. (2024) show that uncertainty-based abstention improves correctness, reduces hallucinations, and increases safety, motivating uncertainty estimation as a deployment-time safety mechanism. African-language evaluation has advanced through human-translated benchmarks, including Uhura for truthfulness and scientific question answering (Bayes et al., 2024) and AfriMMLU within IrokoBench for knowledge evaluation (Adelani et al., 2024), alongside models such as AfroLlama (Jacaranda Health, 2024). However, prior work has focused primarily on capability and benchmark performance. To our knowledge, calibration in African languages, compared across models and paired with mitigation strategies, has not been systematically studied.

3. Methods

3.1. Models

We evaluate four open-weight models chosen to span size, multilinguality, and origin: a small instruction-tuned model, a larger one, a multilingual one, and an Africa-specific one (full specifications in the separately submitted appendix). Open weights give direct log-probability access and full reproducibility without proprietary APIs.

3.2. Datasets

We use Uhura-TruthfulQA (Bayes et al., 2024) (English and six African languages; truthfulness; a variable number of candidate answers per question) and AfriMMLU (Adelani et al., 2024) (18 languages; knowledge; four options per question). Because their language sets and tasks differ, we analyse them separately and never average across them. “African” denotes all languages other than English and French.

3.3. Scoring

For both datasets we score each option by the length-normalised log-probability of its answer text (a cloze reading; detailed in the appendix) and take the softmax over options as the model’s confidence. We deliberately avoid letter-logit scoring (reading the next-token probability of “A”/“B”/“C”/“D”): it silently collapses instruction-tuned models, with Gemma-4 returning “A” for 75% of items and scoring below chance, whereas answer-text scoring collapsed for no model.

3.4. Metrics

We report expected calibration error (ECE, 15 bins), overconfidence (mean confidence minus accuracy), Brier score, error-detection AUROC, and AUARC, the area under the accuracy-rejection curve (definitions in the appendix).

Table 1: English versus African-language performance, by model and dataset. “en” is English and “afr” is the mean over African languages; ECE(TS) is pooled ECE after per-language temperature scaling; AUARC measures abstention quality (higher is better). Models are accurate and calibrated in English but lose both in African languages, and recalibration restores calibration.

Model	Dataset	Acc(en)	Acc(afr)	ECE(en)	ECE(afr)	ECE(TS)	AUARC
Qwen3-4B-Instruct	Uhura-TruthfulQA	0.38	0.32	0.07	0.17	0.03	0.36
Gemma-4-12B	Uhura-TruthfulQA	0.33	0.34	0.21	0.26	0.03	0.39
Aya-Expanse-8B	Uhura-TruthfulQA	0.36	0.31	0.08	0.20	0.04	0.35
AfroLlama-V1	Uhura-TruthfulQA	0.29	0.33	0.18	0.13	0.04	0.35
Qwen3-4B-Instruct	AfriMMLU	0.53	0.30	0.06	0.15	0.02	0.35
Gemma-4-12B	AfriMMLU	0.30	0.27	0.31	0.34	0.07	0.29
Aya-Expanse-8B	AfriMMLU	0.52	0.29	0.06	0.18	0.02	0.34
AfroLlama-V1	AfriMMLU	0.36	0.28	0.22	0.20	0.03	0.31

3.5. Interventions

We evaluate three: per-language temperature scaling, selective abstention by per-language confidence threshold, and 5-shot in-context prompting with examples held out from evaluation (detailed in the appendix).

4. Results

Table 1 summarises every model on both datasets; the central finding is the gap between English and African-language calibration.

1. **Calibration degrades from English to African languages.** Across the models and both tasks, ECE and overconfidence are substantially higher in African languages (Table 1, Figure 1); models are most overconfident where they are least accurate.
2. **The accuracy collapse is a knowledge story.** On AfriMMLU the capable models fall from about 0.52 (English) to about 0.29 (African, near the 0.25 chance floor). On Uhura accuracy is uniformly low (about 0.31) with little English advantage, so Uhura supplies the calibration signal and AfriMMLU the accuracy decay.
3. **Mitigations are unequal.** Per-language temperature scaling cuts pooled ECE to about 0.02 to 0.07 for every model (see appendix); abstention helps less, because uncertainty is a weak error-ranker (AUROC about 0.55), so AUARC sits only modestly above baseline. Few-shot prompting gives small gains for three models and backfires on Gemma (see appendix).
4. **Bigger is not safer.** Gemma-4-12B is the worst-calibrated model (ECE up to 0.34) and is overconfident across every topic, despite being the largest model we evaluate.

Per-language bootstrap 95% confidence intervals are reported in the appendix.

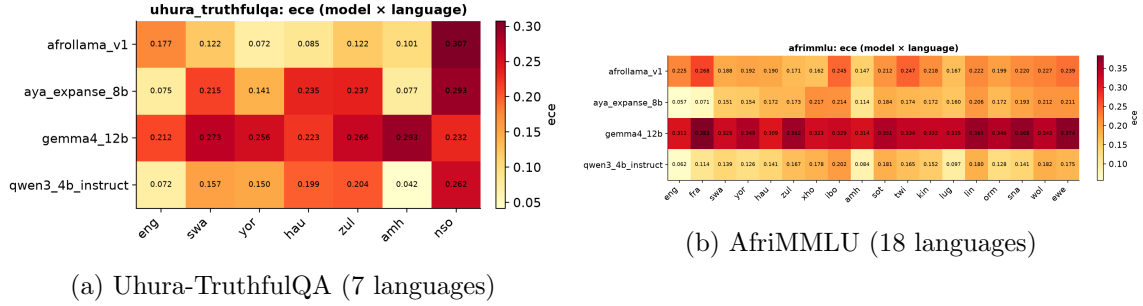


Figure 1: Expected calibration error by model and language. ECE is low in English and French and rises across African languages; Gemma is worst throughout.

5. Discussion and Limitations

Implications for African AI safety. In African languages these models are at once less accurate and more overconfident, the worst combination for trust, since confident wrong answers reach users precisely where verification is hardest. Encouragingly, much of the failure is an output-layer problem: cheap, post-hoc, per-language temperature scaling closes most of the calibration gap with no retraining, so safer deployment is feasible now. Abstention alone is insufficient because the uncertainty signal is weak, so the durable fix remains better African-language data and training. We distil this into a deployment framework (detailed in the appendix): recalibrate, then route on calibrated confidence to answer, ask for more context, or abstain, logging low-confidence queries for data collection.

Limitations. Accuracies are near chance on these hard, low-resource tasks, so we document systematic overconfidence rather than catastrophic confident-wrongness, and absolute ECE is moderate except for Gemma. The weak uncertainty signal (AUROC about 0.55) bounds how much abstention can help. We cover four open models and two benchmarks; reasoning and frontier closed models were out of scope.

Future work. Reasoning-model and closed-model references; in-context calibration at scale; the offline data-collection loop in the deployment framework; and human-grounded evaluation of abstention.

6. Conclusion

Open LLMs are confidently wrong in African languages, less accurate and more overconfident than in English across the models and both tasks. Post-hoc per-language temperature scaling is a cheap, effective safety lever that closes most of the calibration gap, and our deployment framework pairs it with thresholded abstention for safer real-world use.

References

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70:1321–1330. arXiv:1706.04599.

- Kuhn, L., Gal, Y., and Farquhar, S. (2023). Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. *International Conference on Learning Representations (ICLR)*. arXiv:2302.09664.
- Tomani, C., Chaudhuri, K., Evtimov, I., Cremers, D., and Ibrahim, M. (2024). Uncertainty-Based Abstention in LLMs Improves Safety and Reduces Hallucinations. arXiv:2404.10960.
- Bayes, E., Azime, I. A., Alabi, J. O., Kgomo, J., Eloundou, T., Proehl, E., Chen, K., Khadir, I., Etori, N. A., Muhammad, S. H., Mpanza, C., Thete, I. P., Klakow, D., and Adelani, D. I. (2024). Uhura: A Benchmark for Evaluating Scientific Question Answering and Truthfulness in Low-Resource African Languages. arXiv:2412.00948.
- Adelani, D. I., et al. (2024). IrokoBench: A New Benchmark for African Languages in the Age of Large Language Models. arXiv:2406.03368.
- Jacaranda Health (2024). AfroLlama-V1. Hugging Face. https://huggingface.co/Jacaranda/AfroLlama_V1.

Appendix

The appendix (model and setup details, metric definitions, prompts, intervention details, the deployment framework, additional per-language results, reliability curves, and bootstrap confidence intervals) is provided as a separate PDF file, in compliance with the Trustworthy AI Workshop (Deep Learning Indaba 2026) submission guidelines.